

Psychomotor Skills in Medical Ultrasound Imaging

An Analysis of the Core Skill Set

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Sonographers use psychomotor skills to perform medical ultrasound examinations. Psychomotor skills describe voluntary movements of the limb, joints, and muscles in response to sensory stimuli and are regulated by the motor neural cortex in the brain. We define a psychomotor skill in relation to medical ultrasound imaging as “the unique mental and motor activities required to execute a manual task safely and efficiently for each clinical situation.” Skills in clinical ultrasound practice may be open or closed; most skills used in medical ultrasound imaging are open. Open skills are both complex and multidimensional. Visuomotor and visuospatial psychomotor skills are central components of medical ultrasound imaging. Both types of skills rely on learners having a visual exemplar or standard of performance with which to reference their skill performance and evaluate anatomic structures. These are imperative instructional design principles when teaching psychomotor skills.

Key Words—procedural skills; psychomotor skills; skills; sonography; ultrasound; ultrasound education

The skills used to perform a medical ultrasound examination are unique to sonography. Globally, medical ultrasound imaging has been performed for more than 40 years. To date, there has been limited research or literature describing the core psychomotor skills used in medical ultrasound imaging. There is, however, theory and practice evidence of psychomotor skills used in clinical practice in other health professions: for example, nursing and surgical medicine. This literature has informed the analysis of the core psychomotor skills used in medical ultrasound imaging.

For each ultrasound examination performed, fundamental motor skills are used to enable an operator to move and manipulate a transducer in response to sensory information. Explicating and substantiating the theory relevant to ultrasound psychomotor skills provide knowledge and understanding of established and practiced skills. Having established the basic theory and constructs of the psychomotor skills used in medical ultrasound imaging, this knowledge informs clinical practice skill acquisition and skill teaching practice. Application of this knowledge is an important starting point for developing educational approaches that teach scanning skills. This article presents 4 key premises for ultrasound teaching: (1) a description of a psychomotor skill; (2) a proposed definition of a psychomotor skill relevant to medical ultrasound imaging; (3) an analysis of the types of skills used in medical ultrasound imaging; and (4) a brief presentation on how this theory and knowledge may influence psychomotor skill instructional design practice.

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Psychomotor Skills in Ultrasound Imaging

Psychomotor skills describe voluntary movements of the limb, joints, and muscles, in response to sensory stimuli and are regulated by the motor neural cortex in the brain.¹ The literature demonstrates that procedural, technical, or task-based skills are used interchangeably with psychomotor skills.^{2,3} In each health discipline, a psychomotor skill is defined by the unique skills that profession uses. Moreover, definitions vary between disciplines, such as surgical medicine, physiotherapy, and nursing. In surgical medicine, Kovacs² defined a psychomotor skill as “the mental and motor activities required to execute a manual task,” whereas Rose and Best,⁴ coming from a physiotherapy background, further refined the definition to include being “performed correctly, efficiently and safely.” In the nursing literature, Bjørk⁵ stated that a technical skill is a “. . . refined pattern of movement or performance based upon and integrated with the perceived demands of the situation.” Amalgamating these statements, we define a psychomotor skill in relation to medical ultrasound imaging as “the unique mental and motor activities required to execute a manual task safely and efficiently for each clinical situation.” The word “correctly” has been purposefully omitted, as the explicit connotation of the word suggests competence, which is a criterion-based judgment.^{4,6}

Open and Closed Psychomotor Skills

Psychomotor skills can be classified as open and closed skills.^{7,8} Closed skills are methodically executed each time without variation. Examples include hand washing and setting up an amniocentesis or paracentesis equipment tray.⁷⁻⁹ Open skills are more relevant to medical ultrasound imaging; they have a core component that incurs variation each time the skill is executed. This variation may result from a number of factors, which include fetal position or number, bodily habitus, organ orientation and position, pathologic conditions, postsurgical changes, and vascular haemodynamics.¹⁰ One example of an open skill is measuring the crown-rump length of a fetus at 7 weeks' gestation. The core skill involves measuring the maximal bipolar length of the fetal pole. The variable component of the skill results from the position of the implanted gestational sac and is different for every pregnancy. Therefore, the location of the fetal pole within the gestational sac will vary for each pregnancy. The transducer movements required to obtain the longest length of the fetal pole will vary for each pregnancy and will be further modified if the uterus is retroverted. The transducer movements required for this skill cannot be learned by rote, as they vary each time the skill is executed.^{7,8}

The fundamental psychomotor skills used by all sonographers include the following: (1) being able to view 3-dimensional anatomy in real time on a 2-dimensional screen; (2) moving a transducer in multiple planes; (3) scanning an organ in a minimum of 2 orthogonal planes; and (4) depicting the optimal image of a structure for a given clinical scenario. Central to the execution of this skill set is the ability to slide, tilt, rotate, and fan the ultrasound transducer through a structure, using different acoustic windows.¹¹ The transducer movements may be single, sequenced, small, or large, adding a further degree of skill complexity. The psychomotor scanning skills used in medical ultrasound imaging include visuomotor and visuospatial skills.

Visuomotor Skills

Visuomotor skills, or hand-eye coordination skills, evolve from the sonographer having a mental image (visual exemplar) of what the skill should look like for a given clinical scenario.^{12,13} A visual exemplar provides a visual standard of performance and is used as a reference image by sonographers when learning a new scanning skill.^{7,14,15} Visuomotor skills use a continual feedback loop, and the visual exemplar is an essential component. Specific limb motor movements guiding the transducer result in a visual image on the monitor. When the visual image of the structure is not clear, sharp, and representative of the visual exemplar, a complex motor-neural analysis is undertaken in the brain. The current image is compared and contrasted to the visual exemplar. The necessary changes required to achieve an image that matches the standard of the visual exemplar are then computed. A motor response is mapped and instigated in the brain. The positions of the upper limb and joints are changed through muscle activation, which modifies the transducer position, angle, or pressure. This process continues to be updated in real time until a visual standard for the skill is achieved.^{3,4,14} With practice, the motor sequences required to move the probe to achieve the visual standard for a given clinical scenario are learned.^{16,17} The clinical scenario may be further complicated by the patient's body habitus, anatomic variation, pathologic condition, or organ mobility.¹⁰

When learning psychomotor skills, these permutations extrapolate the motor skill learning timeline and make it difficult to learn, replicate, and adapt foundational skills.¹⁸ It is with continued and diligent skill practice that the motor movements and motor plans required to move a transducer for a given clinical scenario are learned and recalled.^{16,19}

The learned motor movements may be single, sequenced, or adaptive, and the process is termed “motor

sequence learning.”¹ An example of a single motor sequence is the sliding of the transducer from the fetal head down to the fetal pelvis to provide the bony anatomic landmarks for locating the femoral shaft. Scanning the supraspinatus tendon of the shoulder is an example of a sequenced psychomotor skill. Large transducer movements are used to locate the tendon and bony anatomy of the humeral head. Small transducer movements are executed to show the fibrillar continuity of the tendon fibers to their point of distal insertion. The transducer is moved in multiple planes. The transducer movements for scanning the supraspinatus tendon include angling and tilting while simultaneously sliding from anterior to posterior and distal to medial, as the tendon is scrutinized for changes in echogenicity, contour, and fibrillar continuity.²⁰ A sequenced psychomotor skill may combine any combination of fine and large transducer movements, whereas an example of an adaptive skill would be rolling a patient into a left lateral decubitus position when a right-sided pelvic collection is noted during an ultrasound examination. The purpose of rolling the patient is to assess whether the fluid is mobile or loculated.

Visuospatial Skills

Visuospatial skills describe the sonographer’s ability to create a 3-dimensional or topographic mental picture of anatomy or an abnormality from 2-dimensional images. The mental picture is constructed from scanning an organ in multiple planes, such as transverse, longitudinal, coronal, radial, antiradial, and oblique planes. The edges of a structure may be further scrutinized by holding the ultrasound transducer in an assigned position while slowly rotating it 360° in real time. This action is only possible when the size of the structure or abnormality is smaller than the beam width of the transducer.^{21,22} Viewing anatomy from multiple planes builds a representative 3-dimensional mental image of a structure, which has height, width, depth, and a relationship with other structures. Visuospatial skills are central to a sonographer being cognizant of the axis, deviation, rotation, and anatomic variation of a structure. These skills alert the sonographer to the anatomic topography, inclusive of anatomic variations. They are contingent on the sonographer having a visual image or exemplar of the expected sonographic anatomy for each anatomically normal structure assessed. An example of a visuospatial skill is the ability to recognize a retroverted uterus from the orientation of the fundal position in the sagittal imaging plane. A perceived aberration in the location or integrity of anatomic structures cues a unique and possibly never rehearsed visuomotor response. Single or sequenced psychomotor skills effect changes in the

transducer position, angle, rotation, and probe pressure. These changes are made to ensure that the parenchyma or abnormality is clearly resolved and the structure is thoroughly interrogated.¹⁵ Thus, when anatomic variation or an abnormality is encountered, visuospatial and visuomotor skills are inextricably linked and form a complex skill set composed of cognitive, motor, temporal, and tactile elements.

Summary and Conclusions

This article presents the core psychomotor skills used when performing an ultrasound examination. The described psychomotor skills are but one component of a clinical ultrasound examination. However, these skills are the foundation of all medical ultrasound examinations.

Visuomotor and visuospatial skills are central components of medical ultrasound imaging. Both types of skills rely on learners having a visual exemplar or standard of performance with which to reference their skill performance and evaluate anatomic structures. These are imperative instructional design principles when teaching psychomotor skills. The psychomotor skills used in medical ultrasound imaging include both complex and open skills. As a result, there are many challenges for those who teach and learn psychomotor skills for medical ultrasound imaging. The limited knowledge and understanding of how single and sequenced psychomotor movements are best learned to execute complex and multidimensional transducer movements for performing rudimentary scanning skills remain an enigma.

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